Problem Set 2

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Contents

Problem 1: *The centralizer of an element.*
Let G be a group and let and let $a \in G$. The centralizer of a is the set of all elements that commute with a:

$$C(a) \coloneqq \{ g \in G \colon ga = ag \}$$

(a) In the group S_5, let $\sigma = (135)$. List all the elements of the centralizer $C(\sigma)$.

Solution:

To find all elements in the centralizer of σ, we must find all elements, $\tau \in S_5$ such that $\sigma\tau = \tau\sigma$.

Helping us in that endeavour is Proposition 2.6., which states that if τ satisfies $\text{supp}(\sigma) \cap \text{supp}(\tau) = \emptyset$, then τ and σ commute. Clearly, σ permutes 1, 3, and 5 while leaving 2 and 4 fixed. So, by definition of the support, $\text{supp}(\sigma) = \{ 1, 3, 5 \}$. Therefore, any τ satisfying $\text{supp}(\tau) \subseteq \{ 2, 4 \}$ must commute with σ. Evidently, there are only two such τ and so we can conclude that at least:

$$\tau = (2, 4),\ \tau = ()$$

must permute with σ and so belong to the centralizer $C(\sigma)$.

We also know, that σ^{-1} must commute with σ (since by definition of inverses, $\sigma\sigma^{-1} = \sigma^{-1}\sigma = 1$). Therefore, seeing as $\sigma^{-1} = (531)$, we can also conclude that:

$$\tau = (5, 3, 1)$$

must also permute with σ, hence making $C(\sigma) \supseteq \{ (), (2, 4), (5, 3, 1) \}$

(b) Prove that $C(a) \leq G$ for all $a \in G$.

Solution:

First, let $a \in G$ be arbitrary. To prove that $C(a) \leq G$, we must show that $C(a)$ is a subgroup of G, which we can do by the one-line subgroup test!

The one line subgroup test states that if $xy^{-1} \in C(a)$ for all $x, y \in C(a)$, then $C(a)$ must be a subgroup of G. So, to that end, letting $x, y \in G$ be arbitrary, I will now proceed to show that $(xy^{-1})a = a(xy^{-1})$, hence meaning that $xy^{-1} \in C(a)$ and concluding the proof.

By definition of $C(a)$, we know that $x, y \in a$ implies that $xa = ax$ and $ya = ay$. We also know, since G is a group and $C(a)$ is clearly a subset of G, that y must have an inverse in G. Letting $y^{-1} \in G$ be y's inverse, we can see that:

$$\begin{aligned}
y^{-1}a &= y^{-1}(a(1)) && \text{(By definition of the identity)}\\
&= y^{-1}(a(yy^{-1})) && \text{(By definition of inverses)}\\
&= y^{-1}((ay)y^{-1}) && \text{(By associativity)}\\
&= (y^{-1}(ay))y^{-1} && \text{(By associativity)}\\
&= (y^{-1}(ya))y^{-1} && \text{(Since } y \in C(a)\text{)}\\
&= ((y^{-1}y)a)y^{-1} && \text{(By associativity)}\\
&= ((1)a)y^{-1}y^{-1} && \text{(By definition of inverses)}\\
&= ay^{-1} && \text{(By definition of the identity)}
\end{aligned}$$

Therefore, $y^{-1}a = ay^{-1}$, implying $y^{-1} \in C(a)$ and thereby making $xy^{-1} \in C(a)$ a corollary because:

$$\begin{aligned}
(xy^{-1})a &= x(y^{-1}a) && \text{(By associativity)}\\
&= x(ay^{-1}) && \text{(Since } y \in C(a)\text{)}\\
&= (xa)y^{-1} && \text{(By associativity)}\\
&= (ax)y^{-1} && \text{(Since } x \in C(a)\text{)}\\
&= a(xy^{-1}) && \text{(By associativity)}
\end{aligned}$$

Immediately follows. Thus, by the one-line subgroup test, $C(a)$ is indeed a subgroup of G.

(c) Two friends, Goofus and Gallant, are arguing about the following claim:

"If $a, b \in G$, then $C(a) \cap C(b) = C(ab)$"

Goofus thinks "$\subseteq$" is true, but Gallant thinks "$\supseteq$" is true. Who is correct? Prove one inclusion, and give an explicit counterexample showing that the other inclusion is false in general.

Solution:

I contend, and will proceed to show, that $\subseteq$ is indeed the only correct inclusion. To do this, I intend to formally prove that $(C(a) \cap C(b)) \subseteq C(ab)$ for any arbitrary $a \in G$, for any arbitrary group G, and provide a specific group G with elements $a, b \in G$ such that $(C(a) \cap C(b)) \supseteq C(ab)$ doesn't hold, making Gallant's inclusion false in general.

(I) Prove that $(C(a) \cap C(b)) \subseteq C(ab)$.

To prove this inclusion, I will show that $x \in (C(a) \cap C(b))$ implies that $x \in C(ab)$. So, let any such x be arbitrary.

By definition of set intersection, we know that $x \in (C(a) \cap C(b))$ implies $x \in C(a)$ and $x \in C(b)$, which by definition of the centralizer, means that $xa = ax$ and $xb = bx$. To prove that $x \in C(ab)$, we need only show that $x(ab) = (ab)x$, however from our preceding observations, we can infer that:

$$\begin{aligned}
x(ab) &= (xa)b && \text{(By associativity)} \\
&= (ax)b && \text{(Since } x \in C(a)\text{)} \\
&= a(xb) && \text{(By associativity)} \\
&= a(bx) && \text{(Since } x \in C(b)\text{)} \\
&= (ab)x && \text{(By associativity)}
\end{aligned}$$

And hence, x belongs to the centralizer $C(ab)$. Therefore, since $x \in (C(a) \cap C(b))$ implies $x \in C(ab)$ for arbitrary x, we have that $(C(a) \cap C(b)) \subseteq C(ab)$, proving the inclusion.

(II) Show $(C(a) \cap C(b)) \supseteq C(ab)$ is false in general.

To show that this inclusion is generally false, we need only observe a single group G with specific elements $a, b \in G$ such that $x \in C(ab)$ but $x \notin (C(a) \cap C(b))$.

Problem 2: *True or false?*
Determine which of the following statements are true. Supply a proof or counterexample as necessary. Note that your examples should be fully justified: you must prove that your example has the necessary properties.

(a) "D_5 has exactly one subgroup of order 5."

Solution:

(b) "Every proper subgroup of A_4 is cyclic."

Solution:

(c) "$\mathbb{Q}$ is a cyclic group under addition."

Solution:

This statement is false.

To demonstrate this, let's assume for the purposes of deriving a contradiction, that $\mathbb{Q}$ is indeed cyclic. This means that there exists some $x \in \mathbb{Q}$ such that $\langle x \rangle = \mathbb{Q}$, or in other words, that for arbitrary $y \in \mathbb{Q}$, there exists some positive integer $n \in \mathbb{N}$ such that $nx = y$.

By definition of rational numbers, we know that $x \in \mathbb{Q}$ if there exists integers $p, q \in \mathbb{Z}$ such that $x = \frac{p}{q}$. However, $y = \frac{p}{2q}$ likewise must belong to $\mathbb{Q}$, since closure of $\mathbb{Z}$ under multiplication implies that $2q \in \mathbb{Z}$, meaning both the numerator and denominator of $\frac{p}{2q}$ are both integers, thus satisfying the definition of rationality.

From this choice of y, it now becomes clear that there exists no $n \in \mathbb{N}$ such that $nx = y$, since such an equality would imply that:

$$\begin{aligned} nx &= y \\ \implies n\frac{p}{q} &= \frac{p}{2q} \\ \implies \frac{np}{q} &= \frac{p}{2q} \\ \implies n &= \frac{1}{2} \end{aligned}$$

Thus preventing n from being a positive integer. Thus, it cannot be the case that $nx = y$ for all $y \in \mathbb{Q}$, contradicting our assumption that $\langle x \rangle = \mathbb{Q}$. Hence, $\mathbb{Q}$ cannot be cyclic under addition.

(d) "Let G be a group. Then $Z(G)$ is cyclic."

Solution:

Problem 3: *The relationship between ord(x) and ord(x^m).*
Let G be a group, and let x be an element of G of finite order $n = \text{ord}(x)$. Let m be an integer which is coprime to n. Prove that $\text{ord}(x^m) = n$.

Hint: You may use the following lemma without proof:
Suppose that a, b, c are three positive integers such that a, b are coprime and a divides bc. Then, a divides c.

Solution:

Let G be a group and let $x \in G$ be such that $\text{ord}(x) = n$. To prove that for coprime $m \in \mathbb{Z}, \text{ord}(x^m) = n$, we need to show that n must be the smallest positive number satisfying $(x^m)^n = 1$.

To do this, I first want to establish that $\text{ord}(x^m)$ is finite to begin with. By definition of the order of an element, we know that $\text{ord}(x) = n$ implies that $x^n = 1$. By use of exponent laws, we therefore have that:

$$\begin{aligned}
(x^m)^n &= x^{mn} && \text{(By result of Exercise 4.1.b)} \\
&= (x^n)^m && \text{(By result of Exercise 4.1.b)} \\
&= 1^m && \text{(Since } x^n = 1\text{)} \\
&= 1 && \text{(Inductively from the definition of the identity)}
\end{aligned}$$

Hence, we know that $\text{ord}(x^m)$ must be less than to equal to n. So, letting $k = \text{ord}(x^m)$ where $k \in \mathbb{N}$, we know by definition of the order of x^m that $(x^m)^k = 1$. By the exponent law referenced earlier, this therefore must mean that $x^{mk} = 1$.

Now, from before, we also know not only that $x^n = 1$, but that $\text{ord}(x) = n$. This enables us to use Theorem 4.3! Theorem 4.3 tells us that since n is the order of x, and $x^{mk} = 1$, then $n|mk$. Or in other words, n must divide mk.

However, by the lemma we were so generously given, we therefore have that n must divide k. This is because n divides mk, where n and m were both assumed to be coprime. It therefore quickly follows that k must therefore be equal to n, since k must be the smallest, positive multiple of n.

Therefore, since $k = \text{ord}(x^m)$ and $k = n$, the order of x^m is n, as required.

Problem 4: *The relationship between ord(x), ord(y), and ord(xy).*
This is a follow-up to a question from the May 27th tutorial.

Let G be a group, and let $x, y \in G$ be elements of finite orders m and n respectively. Suppose that $xy = yx$ and $\langle x \rangle \cap \langle y \rangle = 1$. Prove that:

$$\operatorname{ord}(xy) = \operatorname{lcm}(m, n)$$

Solution:

To prove $\operatorname{ord}(xy) = \operatorname{lcm}(m, n)$, I first seek to establish that $\operatorname{ord}(xy)$ is finite to begin with. As before, this is thankfully quite easy to do.

To start off, recall that $m = \operatorname{ord}(x)$ and $n = \operatorname{ord}(y)$ necessarily implies that $x^m = 1$ and $y^n = 1$. Now, let $L = \operatorname{lcm}(m, n)$. By definition of the lowest common multiple, we know that there exists positive integers $k, l \in \mathbb{N}$ (positive since m, n are the orders of x and y respectively and so must themselves be positive) such that $mk = L$ and $nl = L$ respectively. Furthermore, since x and y were assumed to commutative, we can use exponent rules to see that:

$$\begin{aligned}
(xy)^L &= x^L y^L && \text{(By result of Exercise 4.1.c)} \\
&= (x^{mk})(y^{nl}) && \text{(By definition of } k, l\text{)} \\
&= (x^m)^k (y^n)^l && \text{(By result of Exercise 4.1.b)} \\
&= (1)^k (1)^l && \text{(Since } x^m = 1 \text{ and } y^n = 1\text{)} \\
&= (1)(1) && \text{(Inductively from the definition of the identity)} \\
&= 1 && \text{By definition of the identity}
\end{aligned}$$

Hence, we have both that $\operatorname{ord}(xy)$ must indeed be finite, and that $\operatorname{ord}(xy) \leq L = \operatorname{lcm}(m, n)$. To prove that $\operatorname{ord}(xy) = \operatorname{lcm}(m, n)$, we can proceed by idk contradiction I think ran out of time :/.

Problem 5: *Is a direct product of cyclic groups still cyclic?*

(a) Let A and B be two groups. Prove that if $A \times B$ is cyclic, then so are A and B.

Solution:

Let A, B be arbitrary groups satisfying the above condition. To prove the cyclicity of A and B, we need to prove the existence of two elements I will call $c_1 \in A$ and $c_2 \in B$ respectively such that $\langle a \rangle = A$ and $\langle b \rangle = B$. To aid in this endeavour, let $*$ and $\#$ represent A and B's respective group operators.

We begin by leveraging the cyclicity of $A \times B$. By definition of cyclicity, we know there must exist some $c \in (A \times B)$ such that $\langle c \rangle = A \times B$, which further implies that for every element $(a, b) \in (A \times B)$ (that is, any element in the set $\{ (a, b) \colon a \in A, b \in B \}$), there exists some positive integer $n \in \mathbb{N}$ such that $c^n = (a, b)$. Furthermore, we can easily see that $c = (c_1, c_2) \in (A \times B)$ means $c_1 \in A$ and $c_2 \in B$.

By definition of the direct product, we know that $(c_1, c_2)(c_1, c_2) = (c_1 * c_1, c_2 \# c_2)$ and so for any n:

$$
\begin{aligned}
(a, b) &= c^n && \text{(By definition of cyclicity)} \\
&= ((c_1, c_2))^n && \text{(By definition of } c\text{)} \\
&= \Big(\underbrace{c_1 * c_1 * \ldots * c_1}_{n \text{ times}}, \underbrace{c_2 \# c_2 \# \ldots \# c_2}_{n \text{ times}}\Big) && \text{(By definition of the direct product)} \\
&= ((c_1)^n, (c_2)^n) && \text{(By definition of exponentiation)}
\end{aligned}
$$

And so, since for arbitrary $(a, b) \in (A \times B)$, there exists an $n \in \mathbb{N}$ such that $(c_1, c_2)^n = (c_1^n, c_2^n) = (a, b)$, it therefore follows that ${c_1}^n = a$ and ${c_2}^n = b$. Since a and b were arbitrary, we can therefore deduce that such an n must exist for all possible a and b. Thus, by definition of cyclicity, we therefore have that $\langle c_1 \rangle = A$ and $\langle c_2 \rangle = B$, necessarily making A and B cyclic groups.

(b) Suppose that A, B are finite, cyclic groups of orders m and n respectively, and assume that $\gcd(m, n) = 1$. Prove that $A \times B$ is cyclic.

Solution:

Problem 6: *Let's get working with dihedral groups!*
In this problem, you will be investigating the dihedral group D_{17}: the group of adjacency-preserving permutations of a heptadecagon. You will see that you can work with D_{17} even if you don't know anything about permutations or heptadecagons — all you need is algebra.

There are two key elements of this group: the clockwise rotation r, and a reflection s. They obey the following four algebraic rules:

- *Rule #1:* There are thirty-four elements in D_{17}, split into two types:
  - Seventeen **rotations**: $1, r, r^2, \ldots, r^{16}$; and
  - Seventeen **reflections**; $s, sr, sr^2, \ldots, st^{16}$.
- *Rule #2:* $\text{ord}(r) = 17$.
- *Rule #3:* $\text{ord}(s) = 2$.
- *Rule #4:* $rs = sr^{-1}$.

Answer each question using *only* these four rules. You do not need to prove these rules.

(a) Prove that $r^k s = sr^{-k}$ for all $k \in \mathbb{N}$.

Hint: Induction, with Rule #4 being the base case.

Solution:

To prove the above statement for all $k \in \mathbb{N}$, we may use the principle of mathematical induction. If we are successfully able to prove that $r^1 s = sr^{-1}$, and after assuming that $r^k s = sr^{-k}$ holds, show that $r^{k+1} s = s^{-(k+1)}$ naturally must follow, we can invoke the principle of mathematical induction that $r^k s = sr^{-k}$ must be true for all integers $k \in \mathbb{N}$. Below is my attempt of precisely this.

(I) Prove the base case.

Since:

$$\begin{aligned} r^1 s &= rs && \text{(By definition of exponentiation)} \\ &= sr^{-1} && \text{(By rule \#4)} \end{aligned}$$

We can ascertain that the base case does indeed hold.

(II) State the inductive hypothesis.

Letting $n \in \mathbb{N}$ be arbitrary, we can formulate our inductive hypothesis in terms of n as the statement:

$$r^n s = sr^{-n}$$

And thus, the statement $r^k s = sr^{-k}$ is assumed to be true whenever $k = n$.

(III) Prove the inductive step.

To prove the inductive step, we must show that if the inductive hypothesis holds, $r^{n+1}s = rs^{-(n+1)}$ also holds. Now, by exponent laws we know that:

$$\begin{aligned} r^{n+1}s &= (r^n r^1)s && \text{(By result of Exercise 4.1.a)} \\ &= r^n(r^1 s) && \text{(By associativity)} \\ &= r^n(rs) && \text{(By definition of exponentiation)} \end{aligned}$$

And since $rs = sr^{-1}$ by rule #4:

$$\begin{aligned} &= r^n(sr^{-1}) && \text{(By rule \#4)} \\ &= (r^n s)r^{-1} && \text{(By associativity)} \\ &= (sr^{-n})r^{-1} && \text{(By inductive hypothesis)} \\ &= s(r^{-n}r^{-1}) && \text{(By associativity)} \\ &= sr^{-n-1} && \text{(By result of Exercise 4.1.a)} \\ &= sr^{-(n+1)} && \text{(By distributivity of multiplication over addition)} \end{aligned}$$

We therefore have that $r^{n+1}s = rs^{-(n+1)}$ does indeed hold if the inductive hypothesis holds. By the base case, we know that $r^k s = sr^{-k}$ does hold when $k = 1$. Thus, since $k = 1$ is true, and we know that whenever $r^n s = sr^{-n}$ holds, $n + 1$ likewise holds, we can therefore conclude by the principle of mathematical induction that $r^k s = sr^{-k}$ holds for all $k \in \mathbb{N}$

(b) Show that the elements $r, r^2, r^3, \ldots, r^{16}$ all have order 17, and that the elements $s, sr, sr^2, \ldots, sr^{16}$ all have order 2.

Solution:

(c) Find the conjugacy classes of r and s.

Hint: To find the conjugacy class of r, technically you'd have to compute grg^{-1} for each of the thirty-four elements g in D_{17}. That's a lot of work! Is there a simpler way to handle several cases at once?

Solution:

(d) Show that $\langle r \rangle$ is a normal subgroup of D_{17}.

Hint: There is a useful exercise in the June 3rd tutorial.

Solution:

Problem 7: *The diagonal subgroup*
Let G be a group. The **diagonal** of G is the following subset of $G \times G$.

$$\Delta \coloneqq \{ (x, y) \in G \times G \colon x = y \}$$

(a) Prove that $\Delta \leq G \times G$.

Solution:

To prove this, we can once again use the one-line subgroup theorem! Let $a, b \in \Delta$ be arbitrary. To appeal to the one-line subgroup theorem and conclude $\Delta \leq (G \times G)$, we must show that ab^{-1} must also belong to Δ.

By definition of Δ, we can express $a = (x_1, y_1)$ and $b = (x_2, y_2)$ respectively, where $(x_1, y_1), (x_2, y_2) \in (G \times G)$ such that $x_1 = y_1$ and $x_2 = y_2$. Clearly, $(x_2, y_2) \in (G \times G)$ implies that x_2, y_2 are both elements of G, and since G is assumed to be a group, there must exist inverse elements ${x_2}^{-1}, {y_2}^{-1} \in G$ such that $x_2{x_2}^{-1} = {x_2}^{-1}x_2 = 1 = y_2{y_2}^{-1} = {y_2}^{-1}y_2$.

By definition of the direct product then, we know that $({x_2}^{-1}, {y_2}^{-1}) \in (G \times G)$, and additionally that $({x_2}^{-1}, {y_2}^{-1})$ must be the inverse of (x_2, y_2) since:

$$\begin{aligned}
(x_2, y_2)({x_2}^{-1}, {y_2}^{-1}) &= (x_2{x_2}^{-1}, y_2{y_2}^{-1}) \\
&= (1, 1) \\
&= ({x_2}^{-1}x_2, {y_2}^{-1}y_2) \\
&= ({x_2}^{-1}, {y_2}^{-1})(x_2, y_2)
\end{aligned}$$

By definition of the inverse and since the group operation on $G \times G$ is just the group element of G applied pointwise.

Therefore, since it is now established that $({x_2}^{-1}, {y_2}^{-1})$ is the inverse of (x_2, y_2), and since $b = (x_2, y_2)$, we can thereby conclude that $b^{-1} = ({x_2}^{-1}, {y_2}^{-1})$. Therefore, we know that ab^{-1} is given by:

$$\begin{aligned}
ab^{-1} &= (x_1, y_1)({x_2}^{-1}, {y_2}^{-1}) \\
&= (x_1{x_2}^{-1}, y_1{y_2}^{-1})
\end{aligned}$$

Once again by definition of the direct product's group operation. Recalling now that $x_1 = y_1$ and $x_2 = y_2$ (implying ${y_2}^{-1} = {x_2}^{-1}$ by the uniqueness of inverses), we can further see that ab^{-1} can be further expressed as:

$$\begin{aligned}
ab^{-1} &= (x_1{x_2}^{-1}, y_1{y_2}^{-1}) \\
&= (y_1{x_2}^{-1}, y_1{y_2}^{-1}) && \text{(Since } x_1 = y_1\text{)} \\
&= (y_1{x_2}^{-1}, y_1{x_2}^{-1}) && \text{(Since } {x_2}^{-1} = {y_2}^{-1}\text{)}
\end{aligned}$$

Thus, since it is trivially the case that $y_1{x_2}^{-1} = y_1{x_2}^{-1}$, we know that $(y_1{x_2}^{-1}, y_1{x_2}^{-1}) = ab^{-1} \in \Delta$. Thus, by the one-line subgroup test, we can conclude that Δ is indeed a subgroup of $G \times G$.

(b) Prove that ...

Solution:

Problem 8: *Normal subgroups are friends with everyone*
Let G be a group, and let H and K be two subgroups of G. In this problem, we introduce the following definition: H is said to be friends with K if the product set HK is a subgroup of G. (The definition of "product set" is in the May 20th tutorial).

(a) Give an example of a group G containing two subgroups H and K such that H is not friends with K.

Solution:

(b) Prove that if H is a subgroup of G, then $HH = H$. Conclude that H is always friends with itself.

Solution:

(c) Prove that if H is friends with K, then $HK = KH$. Conclude that K must be friends with H too.

Solution:

(d) Suppose that H, K, and L are three subgroups of G. Suppose that H is friends with K, and K is friends with L. Does it follow that H is friends with L too? Either prove it or provide an explicit counter-example.

Solution:

(e) Let N be a subgroup of G. Prove that if N is a normal subgroup of G, then N is friends with every subgroup of G.

Solution: